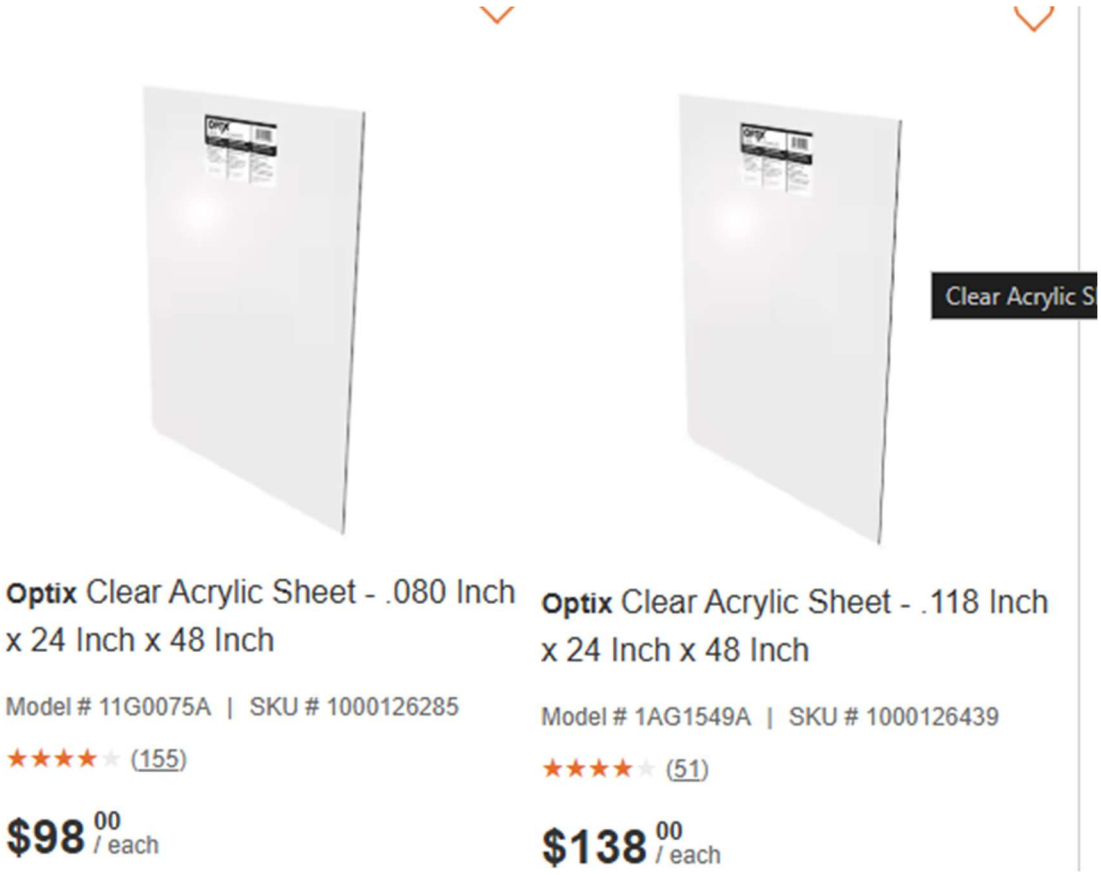
Rail Types

Clear Acrylic Sheets:

https://www.homedepot.ca/en/home/categories/building-materials/plexiglass-and-acrylic-sheets.html

Clear acrylic sheets come in many sizes; the sizes that are both in our budget big enough for our rail system are 0.08x24x48 inch and 0.118x24x48 inch sheets. These acrylic sheets are easily available at local hardware stores.

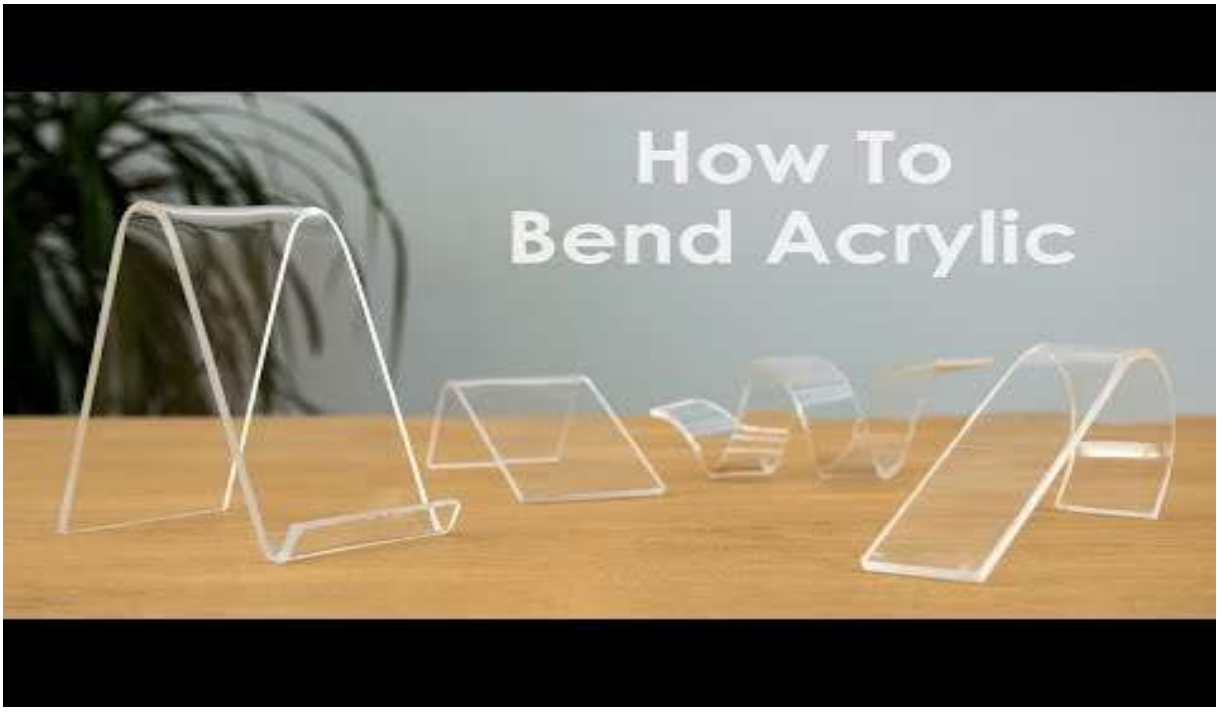


Bending

The best way to fabricate our rails from acrylic is to bend it into the shape using heat. This allows us to get the shape of our rail without having to use fasteners, which would add weak points and potentially interfere with the launch of our UAV. However, bending acrylic can be difficult to execute properly, with high chance of error the cost of this option increases with each attempt. These links show and explain how to properly bend acrylic. A jig would be required in order to ensure consistent and proper shape.

https://uvacrylic.com/blog/bend-acrylic-sheet/

https://www.youtube.com/watch?v=s5nP7MDf63Y&t=1s



Laser Cutting

Acrylic sheets are able to be laser cut allowing us to accurately cut our sheets the desired size for our rail as well as cut out any water drainage before shaping the sheets. Laser cutting is available in the ILC for a cost.

Cost:

One sheet of 24x48inch acrylic should be enough to fabricate both rails not accounting for error. Sheets of 0.08- and 0.118-inch thickness cost 98 and 138 dollars respectively. In addition, the cost of laser cutting is added to the total cost. This method of fabrication is bound to have errors during manufacturing, increasing the total cost exponentially each time.

Pros

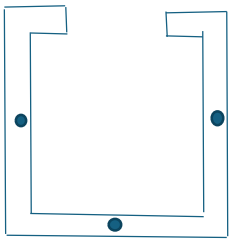
- The molded acrylic allows for the rail to be constructed of one continuous piece
- The fiction within the rail is lower with the smooth acrylic
- Laser cutting allows for easy addition of water drainage holes

Cons

- Difficult to fabricate properly
- High likelihood of error
- Expensive in the inevitable event of errors

3D Printed PLA:

The rail would be printed in sections with dowel holes allowing each segment to be attached together. A rough cross-sectional image is shown below.



This method would require each rail to consist of multiple parts. This adds seems at the connection points where the launch could potentially get caught up on, interfering with the launch. In addition, each seem is a weak point where stress from the force exerted by the launch or the waves could end in structure failure.

This method provides our team with the greatest freedom in our design, allowing us to make the exact shape we want. However, with this method there is the issue of overhang while printing; this will require us to manually drill the drainage holes after printing is complete.

PLA has a tensile strength of 50-60 MPa (https://formlabs.com/blog/strongest-3d-printing-materials/) compared to a tensile strength of 65 MPa and flexural strength of 90MPa for acrylic (https://www.mcneallplastics.com.au/news/everything-about-acrylic-plastics). The layers of 3D printed PLA lead to weak points in the layers when bending.

Cost

https://store.qkits.com/1-75mm-pla-filament-translucent-1-kg.html

3D printing filament is roughly 30 dollars a roll. Assuming that we would need 3-4 rolls to construct the rails with room for error and redesign the total cost would be 90-120 dollars.

Pros

- Customizability
- The costs are lower than that of acrylic
- Ease of fabrication

Cons

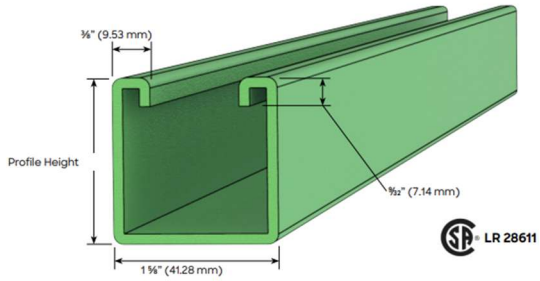
- Multiple sections increasing the number of week points and catch hazards.
- 3D Printed PLA even at 100% fill is the weakest option, with the lowest tensile and flexural strength.
- Higher friction compared to acrylic

Atkore Sasco Strut:

https://images.salsify.com/images/vq6xjhsdelc2ar6clt3e/Sasco%20Strut%20Product%20Catalogue.pdf

S2 SL2-G 120

S2





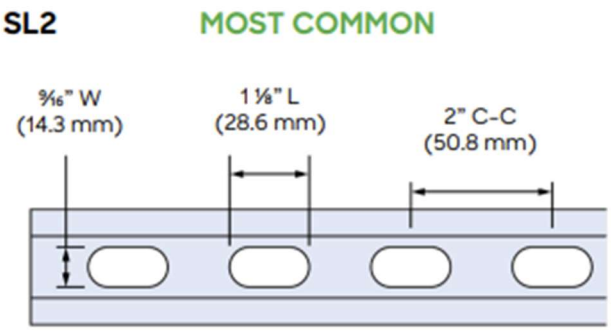
Height: 1-5/8 in

Width: 1-5/8 in

Steel Thickness: 12 gauge

Using these channels as the base of our rails allows the rails to be made from one solid piece without the need for as much fabrication as using acrylic or PLA 3D printing. In addition, these channels provide us a greater amount of stability compared to the other options due to it being made from solid steel. This stability increase allows our system to function more consistently.

Slotted:



The slots in the strut in our case allow the water trapped inside the rail to have places to run out, preventing the rail from trapping an excess amount of water.

-G

These struts are made from galvanized steel; this is due to what is available to us within our budget. It would be much better if we were able to use stainless steel instead due to its improved corrosion resistance as well as its higher tensile strength of 515- 1300 MPa compared to 400-841MPa for galvanized steel.

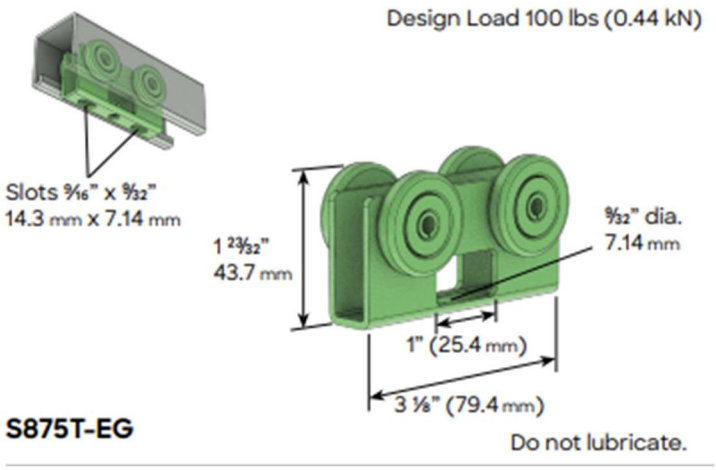
https://www.unifiedalloys.com/blog/stainless-steel-vs-galvanized/

120

This states that the strut is 120 inches long or 10 feet; this allows us to have plenty of room to cut our rails to the appropriate size as well as having excess in the case of error.

Atkore Sasco Strut S875T-EG Trolley:

https://www.atkore.com/en-ca/product/s875t-eg



The S875T-EG trolly is designed to be seamlessly compatible with the S2 strut. The structure of the trolly is made out of electrogalvanized steel with stainless steel wheel bearings that require no lubrication. In addition to working seamlessly with the struts these trolleys allow us to have a solid and stable base to attach our carriage system as well as deliver the force required to launch.

Cost

The S2 SL2-G 120 strut channel and the S876T-EG trolley are typically sold from industrial supply distributors, and individual pricing is hard to obtain. In our case, a group member has access to excess from work costing our team nothing.

Pros:

- These channels are the most structurally sound option
- There is less fabrication in order to implement this into our design, all that is required is to cut them to the propper size.
- There are trolleys specifically designed for these channels
- Group member has access to supply for minimal to no cost
- Come slotted allowing for water drainage

Cons:

- These channels being made from steel weigh more than the other options which in turn puts a greater burden on the angle control actuators.
- Galvanized steel; the galvanized steel will over time corrode and rust compared to if stainless steel were to be used.
- Higher friction compared to acrylic